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Division: A2

ASSIGNMENT NO: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement: Write a Java program to demonstrate the use of the super keyword in inheritance.

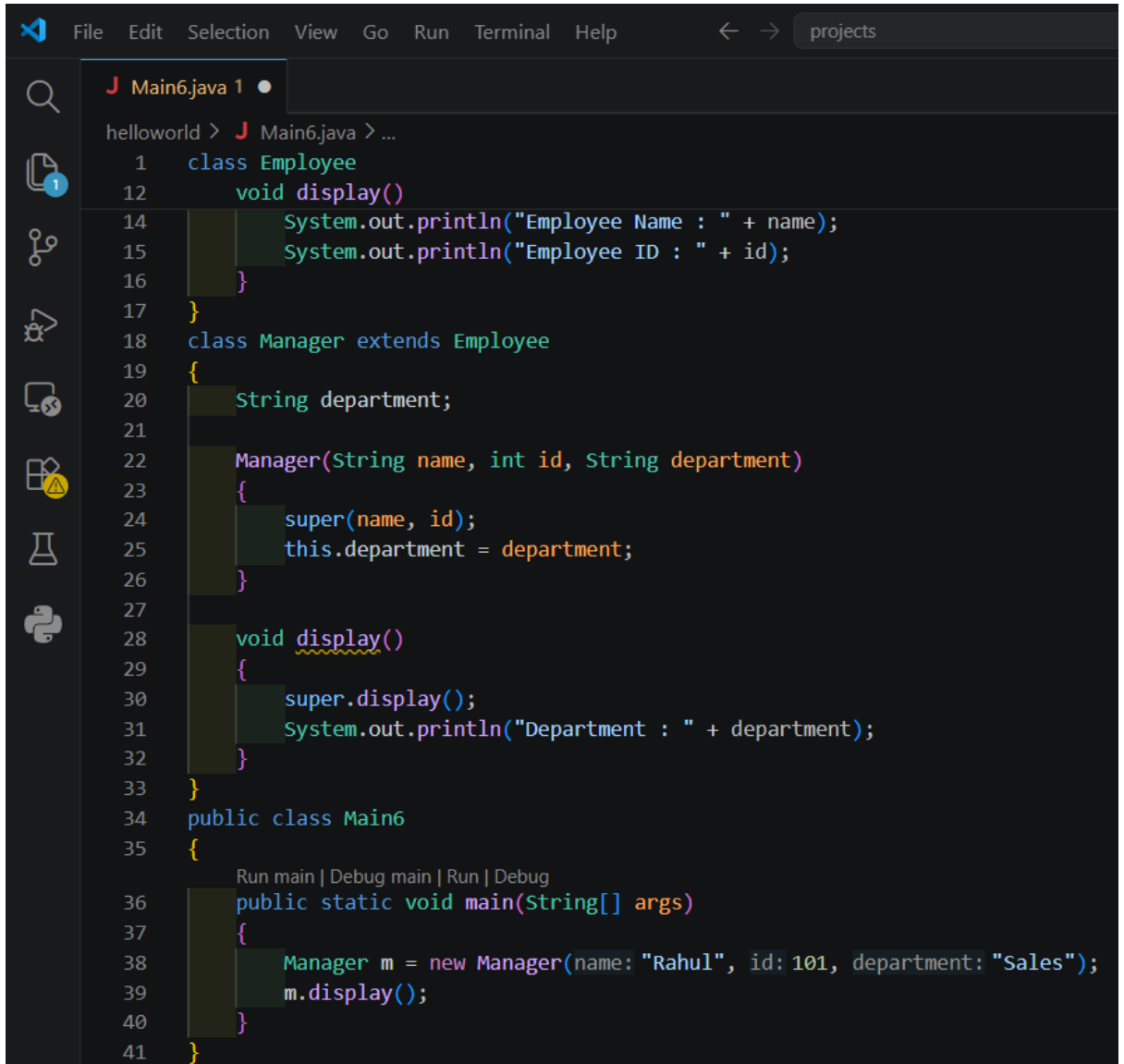
Learning Objectives: To demonstrate the use of the super keyword for accessing parent class members.

Theory : The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Exercise

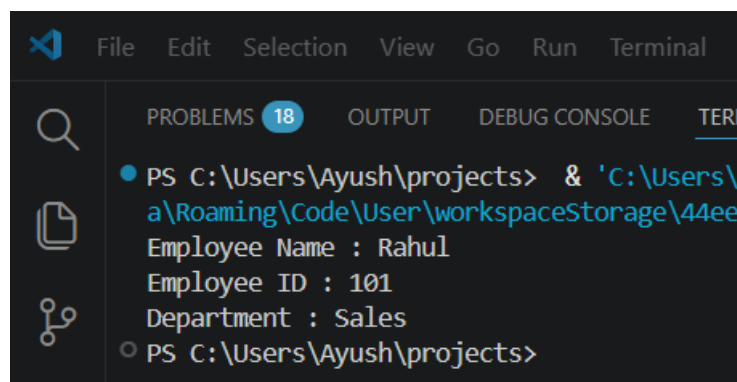
1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Program:



```
File Edit Selection View Go Run Terminal Help projects
Main6.java 1
helloworld > J Main6.java > ...
1  class Employee
12 void display()
14      System.out.println("Employee Name : " + name);
15      System.out.println("Employee ID : " + id);
16  }
17  }
18  class Manager extends Employee
19  {
20      String department;
21
22      Manager(String name, int id, String department)
23      {
24          super(name, id);
25          this.department = department;
26      }
27
28      void display()
29      {
30          super.display();
31          System.out.println("Department : " + department);
32      }
33  }
34  public class Main6
35  {
36      Run main | Debug main | Run | Debug
37      public static void main(String[] args)
38      {
39          Manager m = new Manager(name: "Rahul", id: 101, department: "Sales");
40          m.display();
41      }
42  }
```

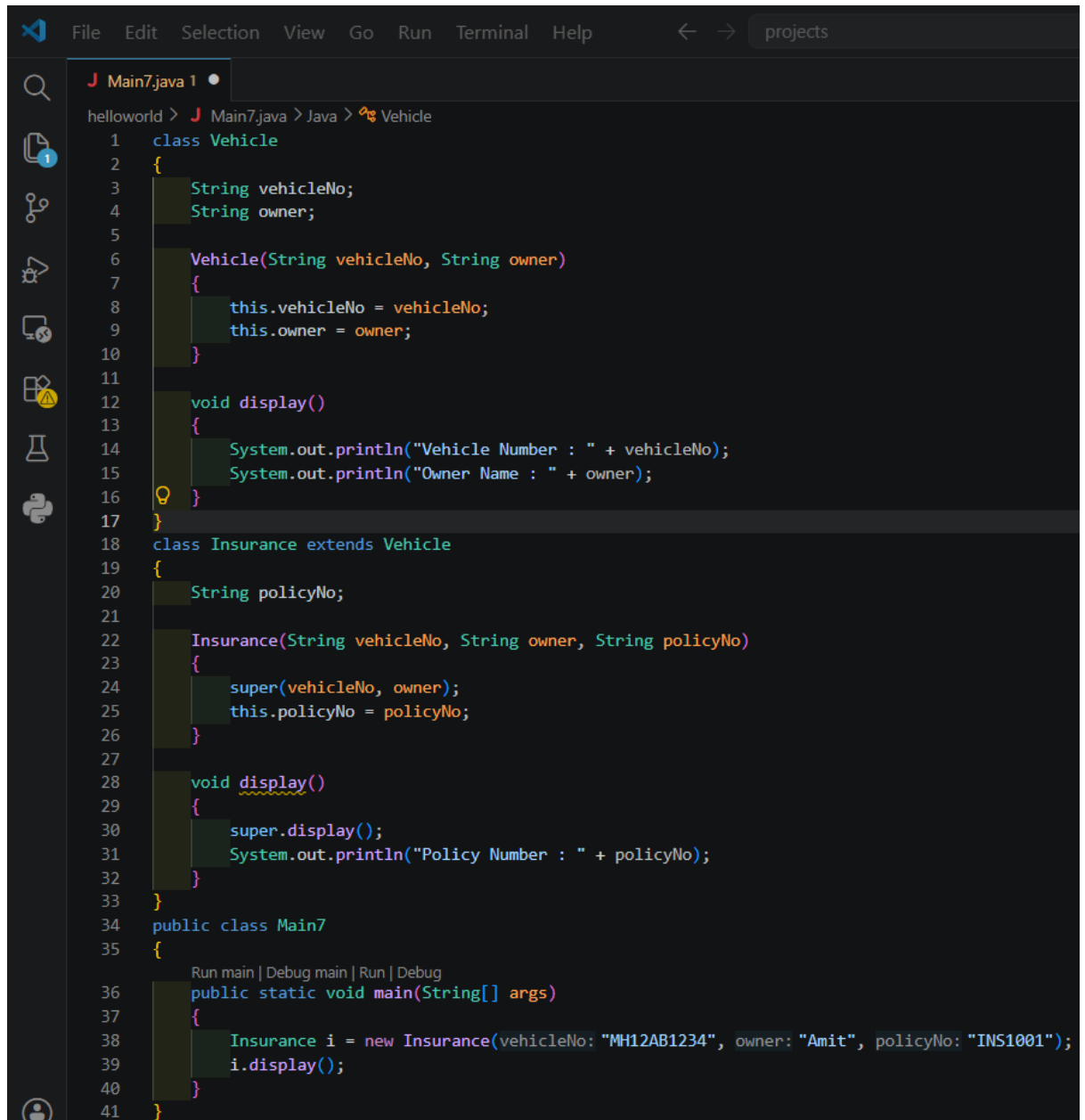
Output:



```
File Edit Selection View Go Run Terminal
PROBLEMS 18 OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\Ayush\projects> & 'C:\Users\Ayush\AppData\Roaming\Code\User\workspaceStorage\44ee...
Employee Name : Rahul
Employee ID : 101
Department : Sales
PS C:\Users\Ayush\projects>
```

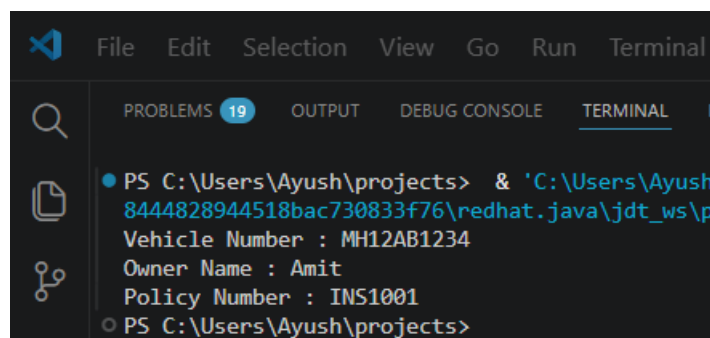
2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Program:



```
1 class Vehicle
2 {
3     String vehicleNo;
4     String owner;
5
6     Vehicle(String vehicleNo, String owner)
7     {
8         this.vehicleNo = vehicleNo;
9         this.owner = owner;
10    }
11
12    void display()
13    {
14        System.out.println("Vehicle Number : " + vehicleNo);
15        System.out.println("Owner Name : " + owner);
16    }
17 }
18 class Insurance extends Vehicle
19 {
20     String policyNo;
21
22     Insurance(String vehicleNo, String owner, String policyNo)
23     {
24         super(vehicleNo, owner);
25         this.policyNo = policyNo;
26     }
27
28     void display()
29     {
30         super.display();
31         System.out.println("Policy Number : " + policyNo);
32     }
33 }
34 public class Main7
35 {
36     Run main | Debug main | Run | Debug
37     public static void main(String[] args)
38     {
39         Insurance i = new Insurance(vehicleNo: "MH12AB1234", owner: "Amit", policyNo: "INS1001");
40         i.display();
41     }
42 }
```

Output:



```
PS C:\Users\Ayush\projects> & 'C:\Users\Ayush\8444828944518bac730833f76\redhat.java\jdt_ws\pr
Vehicle Number : MH12AB1234
Owner Name : Amit
Policy Number : INS1001
PS C:\Users\Ayush\projects>
```